



OPEN Exploring associations between energetic and anthropometric characteristics with front crawl technique in young male swimmers using IMU sensors: a preliminary study

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The study explored the relationships between hand movement patterns and overall kinematic indicators of front-crawl sprint technique in boys aged 12–14, while considering the role of the morpho-functional profile. The sample consisted of 41 participants. Data were collected from anthropometric measurements (8 variables), a 20-s arm-cranking test (6 variables), and a 25-m front crawl trial. Swimming motion was characterized using inertial measurement unit sensors (IMU; BlueTrident) and Artificial Intelligence (8 variables). Video analysis was used to determine swimming velocity (v), stroke rate (SR), stroke length (SL), and stroke index (SI). Principal Component Analysis identified five components: Large Body Size (LBS), Low Fat (LF), Enormous Power (EP), Dynamism (D), and High Angular Velocity (HAV). Pearson correlation analysis revealed strong associations between LBS and EP ($r = 0.85$), as well as between D and EP ($r = 0.51$) and D and LBS ($r = 0.34$). Velocity was related to LBS, EP, D, and LF ($r = 0.38–0.56$). SR showed strong associations with D ($r = 0.76$) and EP ($r = 0.35$), whereas SL was negatively associated with D ($r = -0.44$). SI was related to LBS, LF, and EP ($r = 0.38–0.49$). Overall kinematic indicators are associated with hand acceleration and angular velocity, and IMU based technique assessment supports evaluation of how effectively young swimmers utilize their anthropometric–energetic potential.

Keywords Swimming, Kinematics, Anthropometry, Mechanical muscle power, Youth

Due to its complex structure, a swimming race is typically divided into zones: start, turn (if present), and full-stroke (also called “clean swimming”) ^{1,2}. The efficiency of movement in each zone is determined by various factors ³, placing different demands on the athlete’s motor skills and physical fitness ⁴. For instance, full-stroke swimming velocity depends on the swimmer’s capacity to generate substantial propulsive force using the upper and lower limbs while simultaneously minimizing hydrodynamic drag ⁵. The integration of these determinants reflects high-quality movement execution, which, in turn, is unattainable without an appropriate somatic structure, optimal physical fitness, and technical skills. Proper movement execution is especially critical in sprint events (50–100 m), where even small technical errors may substantially reduce the likelihood of competitive success ⁶.

Full-stroke technique is typically described using overall kinematic parameters such as velocity (v), stroke rate (SR), stroke length (SL), and stroke index (SI; velocity $\times$ stroke length). These indices only provide a general description of the athlete’s movement pattern. However, as Morais et al. ⁷ point out, the proper combination of SR

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and SL is crucial for fast swimming. SR and SL values are typically determined based on video analysis of recorded footage⁸. Recent technological developments allow for more detailed assessments using alternative methods. One such example involves inertial measurement units (IMU), which, due to their waterproof construction, are increasingly used in swimming research⁹. IMU sensors are compact devices containing accelerometers and gyroscopes that enable, among other functions, the measurement of translational acceleration and rotational velocity¹⁰. A major advantage of IMU technology is its ability to capture movement patterns of specific body segments (e.g., upper limbs), including intra-cycle variability such as fluctuations in angular velocity or acceleration^{11,12}.

Commercially available IMU systems designed specifically for swimming have recently entered the market¹³. These devices are typically accompanied by software capable of comprehensive data processing, enabling advanced visualization and the computation of additional movement metrics, such as temporal changes in hand velocity during the underwater phase¹⁴. However, such solutions often exceed the financial capabilities of most training centers, particularly those working with youth athletes¹². In recent years, however, an alternative to commercial technique analysis systems has emerged in the form of movement recording using less expensive IMU sensors and data processing supported by artificial intelligence (AI)^{10,15}. Similar approaches have been successfully adopted in various non-sport domains¹⁶, and their application in sport holds considerable promise for extending advanced technique analysis beyond the highest-level groups¹³.

It is well established that swimming technique is influenced by numerous individual characteristics, among which physical fitness and somatic structure are most frequently highlighted^{7,17}. In sprint events (50–100 m), it has been well documented that larger athletes with high anaerobic power achieve better results¹⁸. Nevertheless, many previous studies examining determinants of performance have not sufficiently addressed how these characteristics interact with swimming technique, including their influence on SR and SL.

The aim of the present study was to examine the associations between hand movement patterns and overall kinematic indicators of front-crawl sprint technique in boys aged 12–14. To achieve this, an analytical approach was applied, combining IMU-based measurement of hand motion with AI-based data processing. The relationship between technical and biological factors was examined by characterizing the athletes' morpho-functional profiles using Principal Component Analysis (PCA). A secondary objective was to evaluate the associations between front-crawl overall kinematic parameters - velocity, Stroke Length, Stroke Rate, and Stroke Index - and PCA-derived components representing anthropometrics, upper-limb mechanical power, and variables describing upper-limb motion.

Methods

Study group

The initial study group consisted of 47 male swimmers aged 12–14 years. Subjects were volunteers recruited from among the best regional swimmers in their age categories selected for the District Team. Participants were active swimmers with a valid national swimming federation license who competed in freestyle events, performing front crawl technique. Each athlete had 4–6 years of training experience. The average score in the 100 m freestyle (short-course swimming pool) achieved by the athletes within six months prior to the study was 311 ± 77 World Aquatics points. According to the classification by Ruiz-Navarro et al.¹⁹, three participants represented level 4 (above 450 World Aquatics points), while the remaining represented level 5 (below 450 World Aquatics points).

The study was approved by the Regional Medical Chamber in Krakow (no. 18/KBL/OIL/2025). All subjects and their parents were informed about the research procedure before the study and provided informed consent for their participation. Subjects were instructed to withdraw from the study at any stage. All procedures contributing to the study complied with the ethical standards of the relevant national and institutional committees on human experimentation and the Declaration of Helsinki of 1975, revised in 2008 as well as good clinical practice guidelines.

The study consisted of three stages: two conducted on land and one in the water. Anthropometric measurements were taken on land, followed by a test to assess the mechanical power of the upper limbs (arm-cranking). The in-water stage involved participants completing a 25-meter front crawl trial, during which the movement of the upper limbs (using IMU sensors) and the whole body (video capturing) were recorded.

Anthropometric measurements

Anthropometric data were collected based on the methodology of Martin and Saller²⁰ and Tanner²¹. The measurements were taken by two qualified individuals with appropriate knowledge and experience in the field of anthropometry. Length measurements were made using an anthropometer, a small spreading calliper from the Sieber Hegner Machines SA set (GPM, Switzerland), and non-stretchable tape (accuracy of 1 mm). Skinfold thickness was measured using a Holtain calliper (GPM, Switzerland) with an accuracy of 0.5 mm. An electronic scale, Tanita BC-418 (Japan), with bioelectrical impedance measurement, was used to determine the mass and composition of the body and its segments (with an accuracy of 0.01 kg). When necessary, measurements were taken on the right side of the body. Based on the measurements, the following were determined:

- body height – H [cm];
- surface of the hand – S_H [mm²]: calculated as: $S_H = 3.14 \cdot \frac{WH \cdot LH}{10}$, where: WH (width of the hand) – as the distance between the metacarpale radiale and metacarpale ulnae points; LH (length of the hand) – as the distance between the stylium and dactylium points²²;
- F_T : fold on triceps [mm];
- F_S : fold on shoulder blade [mm];

- S_A : arm span [cm] - as the distance between the dactylion points of the right and left hand, with the arms extended in a standing position;
- M : body mass [kg];
- F_B : body fat [%];
- A_{MB} : active tissue mass of body [kg];
- A_{MUL} : active tissue mass of upper limbs [kg];
- A_{MT} : active tissue mass of trunk [kg].

Mechanical power measurements

To assess the mechanical power of the upper limbs, a 20-second maximal test was performed on an upper limb ergometer (834E-Ergomedic, Monark, Sweden). The testing procedure was fully consistent with that described by Wadzyk et al.⁸. The load on the ergometer was 3% of the subject's body mass.

Each subject underwent a short, several-minute general and specific warm-up (to familiarize them with the nature of the test). The subjects were tasked with performing 20 s of arm-cranking at the highest possible intensity. The start and end of the task were signaled by the person performing the measurement.

The ergometer was operated using the MCE program (version 5v2, JBA Staniak, Poland), which allowed for the determination of the following variables:

- P_p : peak power [W];
- P_{PREL} : relative peak power - normalized to body mass [W/kg];
- P_M : mean power [W];
- P_{MREL} : relative mean power - normalized to body mass [W/kg];
- T_{PP} : time to reach peak power – from the start of the test to the development of peak power [s];
- F_I : Fatigue Index – calculated as: $FI = \frac{WND}{WD} \cdot 100\%$, where: WD – work that the athlete would perform if, after reaching P_p , they developed maximum power until the end of the test; WND – the difference between WD and the work that the subject actually performed.

Each participant completed a single maximal trial, consistent with the cross-sectional design of the study. Accordingly, test–retest reliability (ICC) was not evaluated within this cohort.

Registration of front crawl technique

The measurements were conducted in a 25-meter swimming pool. The test was preceded by a general warm-up on land and a specialized (in-water) warm-up. The in-water preparation included 300 m of low-intensity effort (100 m front crawl, 100 m backstroke, 100 m front crawl drills), 4 repeats of 50 m front crawl, 15 m of which were completed at maximum intensity (the remaining 35 m at low intensity), and 300 m of low-intensity effort (structured as previously described). All repetitions were separated by approximately a 30-second rest. The intensity was controlled by the swimmers, however the exercises were supervised by a certified swimming coach. After an approximately 5-minute rest and familiarization with the task, the participants swam 25-meter front crawl at their maximum speed. Participants began the task in the water by pushing off from the starting wall without performing underwater dolphin kicks. Breathing was allowed for the first 10 m after the start, but participants were required to refrain from inhaling during the recording zone (the next 10 m).

The movement course was recorded using a GoPro Hero Black 7 camera (GoPro Inc., USA) in the “Linear” video recording mode at a frequency of 60 frames/s, positioned perpendicular to the subject's direction of movement. The device was placed approximately 5 m above the pool, approximately 8 m from the lane where the participants were swimming, and 15 m from the starting wall. This enabled the recording of a 10–20 m zone previously marked with black tape on both sides of the pool.

The kinematic variables describing the overall technique were determined according to the methodology of Wadzyk et al.⁸. The mean values were determined using Kinovea software (version 0.8.15, Joan Charmant & Contrib, France):

- v : velocity [m/s];
- SR : Stroke Rate [cycle/min];
- SL : Stroke Length [m];
- SI : Stroke Index [m²/s].

During the test, BlueTrident sensors (VICON, United Kingdom) were placed on the subjects' hands, approximately 3 cm above the radiocarpal joints. The wristbands carrying the devices were additionally secured with elastic tape to prevent movement of the sensors relative to the subject's body. This sensor placement enabled the acquisition of kinematic data describing upper-limb movement patterns, which represent the primary source of propulsion in front-crawl swimming. The sensors were operated using an iPad (version 9, Apple, United States) with Capture.U software (version 1.1.1.993, VICON, United Kingdom) installed. The IMU sensors were calibrated in advance according to the user manual provided with the devices.

Raw data was exported as a CSV file to Excel using Capture.U Desktop software (version 1.4.1, VICON, United Kingdom). The data, sampled at 60 Hz, contained values for translational acceleration and angular velocity in three axes. Measurements taken in the first 5 and last 3 s of the test were eliminated from the data. This resulted in minimum 5–7 s of recording, containing a minimum of 3 movement cycles for each subject.

A semi-automated, rule-based analytical pipeline incorporating ChatGPT-4 (OpenAI, San Francisco, CA, USA) was implemented for further data analysis (Fig. 1). The language model did not autonomously perform analyses but supported predefined computational steps under researcher supervision.

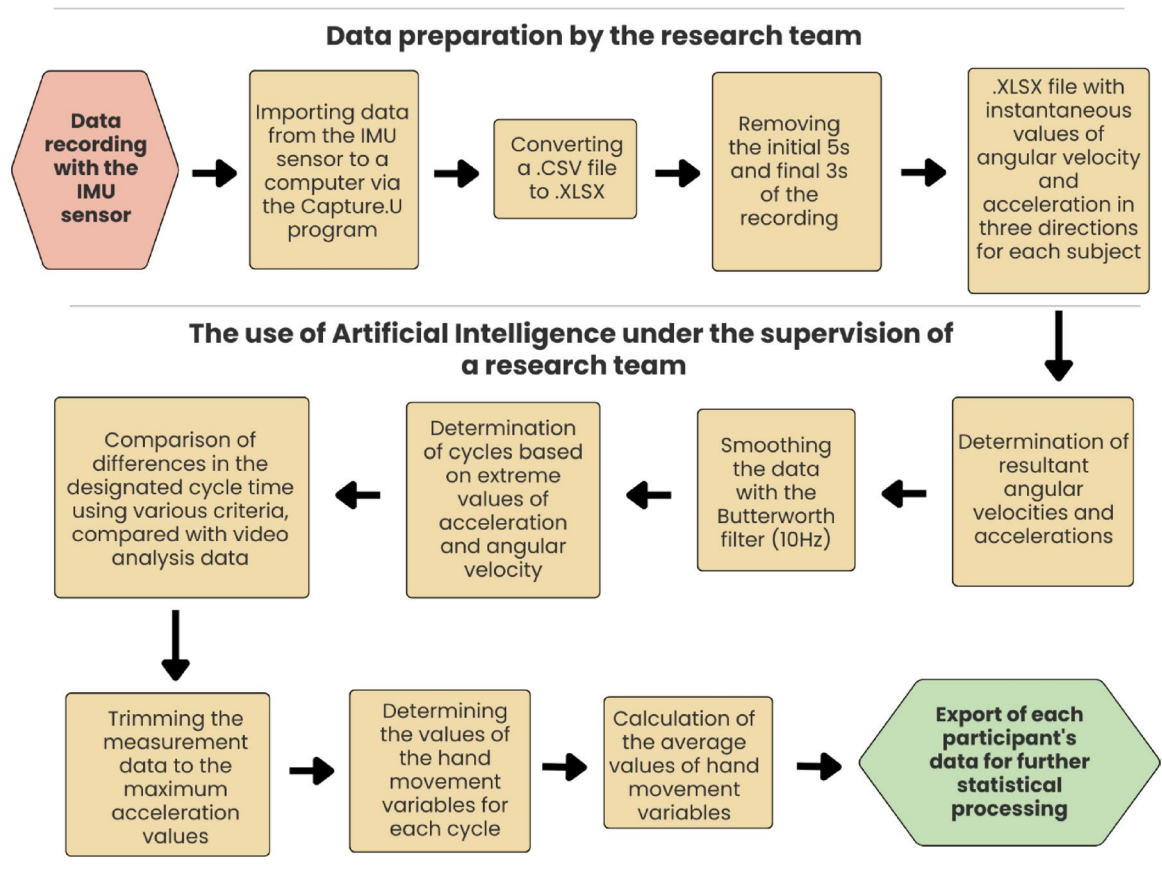


Fig. 1. Schematic representation of the data processing workflow performed by the research team with the support of artificial intelligence.

First, data from three directions were used to determine the resultant accelerations and angular velocities and then filtered using a Butterworth filter (10 Hz). This preprocessing step was identical for all participants and trials. The researcher verified the correctness of data filtering by comparing plots of the raw and filtered signals. The next step was to determine full cycles based on the identification of minimum and maximum values for angular velocity and accelerations. It was assumed by research team that the cycle duration cannot be shorter than 80% of the value determined based on the video analysis. Based on this, the average duration of the distinguished cycles was determined for each subject and compared with the corresponding average time calculated based on video analysis. The maximum allowable difference between the two methods of determining cycle time was set at 4%²³. In isolated cases where this threshold could not be met, the initially identified cycles were visually inspected to assess potential edge-related artifacts (Fig. 2). Based on this inspection, the researcher evaluated whether excluding the first and/or last detected cycle improved agreement with video-based cycle duration estimates, while maintaining the same cycle-definition criterion (Fig. 3). The option yielding a smaller discrepancy relative to video analysis was retained; otherwise, the original cycle set was preserved.

The average cycle duration determined based on peak acceleration values in the study group differed by 1.06% from the time obtained from video recordings; in one isolated case, the difference exceeded the 4% threshold (4.76%). These were the smallest differences compared to other criteria (acceleration minimum and angular velocity extremes), for which the values observed were within the range of 1.2–1.5%. Therefore, movement cycles were identified based on acceleration peaks. For the selected approach, agreement between IMU- and video-derived cycle duration was assessed using the intraclass correlation coefficient (ICC; two-way random effects, absolute agreement, single measurement), yielding a value of 0.99.

Using the identified cycles, the average course of angular velocity and acceleration changes within each cycle was determined for each participant. Subsequently, the following were determined:

- ACC_{MEAN} : mean acceleration [m/s^2].
- ACC_{MAX} : maximum acceleration [m/s^2].
- ACC_{MIN} : minimum acceleration [m/s^2].
- ACC_{RMS} : rms acceleration - root mean square of acceleration [m/s^2].
- $GYRO_{MEAN}$: mean angular velocity [deg/s].
- $GYRO_{MAX}$: maximum angular velocity [deg/s].
- $GYRO_{MIN}$: minimum angular velocity [deg/s].

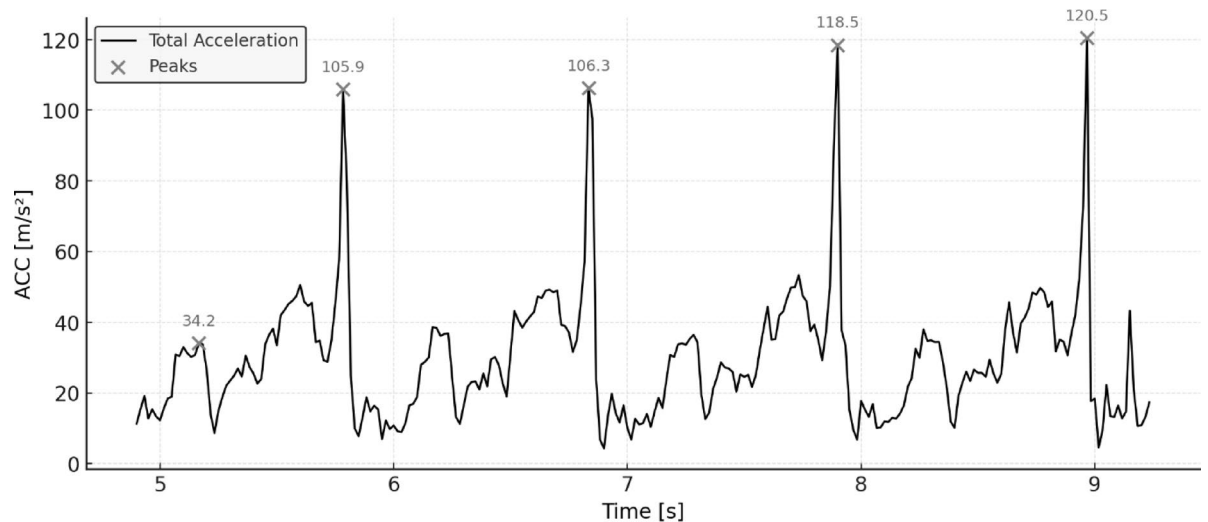


Fig. 2. An example of incorrect identification of peak angular acceleration values used to determine full motion cycles.

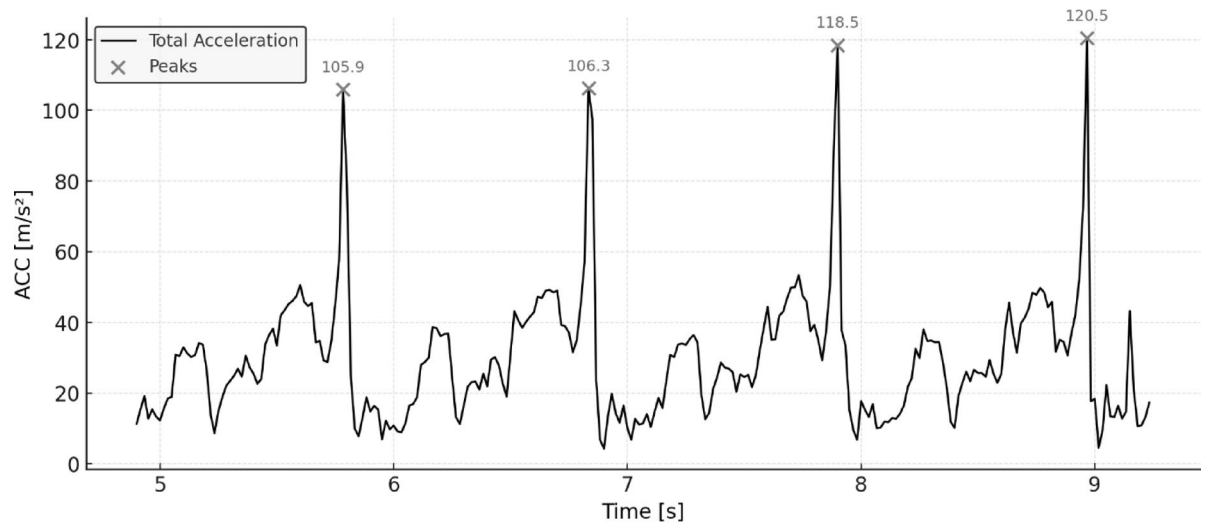


Fig. 3. An example of proper identification of peak angular acceleration values used to determine full motion cycles.

– $GYRO_{RMS}$: root mean square of angular velocity [deg/s].

Statistical procedure

Statistical analysis was performed in Statistica (version 13, StatSoft, Poland). Data were grouped into blocks: anthropometrics, mechanical power, upper limb movement (IMU sensors) and those describing the overall swimming technique (video recording). Descriptive statistics (mean $\pm$ SD) were calculated for all indices.

For the last category of variables, Pearson r correlation coefficients were calculated between v , SR, SL, and SI and were reported together with sample size and 95% confidence intervals (CI). Correlation analyses were exploratory and descriptive in nature. Therefore, no formal hypothesis testing or multiple-comparison correction was applied, and results were interpreted based on effect sizes (ES) and confidence intervals rather than statistical significance. Pearson correlation coefficients were interpreted as measures of ES. The magnitude of the correlations was classified using the following thresholds:

$|r| < 0.1$ - trivial;
 $0.1 \leq |r| < 0.3$ - small;
 $0.3 \leq |r| < 0.5$ - moderate;
 $0.5 \leq |r| < 0.7$ - large;
 $0.7 \leq |r| < 0.9$ - very large;
 $0.9 \leq |r| < 1.0$ - nearly perfect²⁴.

Variables from the first three groups were standardized, and then, using the Mahalanobis distance²⁵, potential outliers were identified by block division. Based on this, six participants were classified as outliers. After removing them from the study group and recalculating the Mahalanobis distance, no outliers were identified. All subsequent statistical analyses were therefore performed on a final sample of 41 participants.

The next step was to perform a PCA, dividing them into blocks: anthropometric, mechanical power, and upper limb movement. Each data category was analysed separately. Number of factors was determined based on eigenvalues and the Kaiser criterion²⁶. This resulted in two anthropometric factors, one mechanical power factor, and two upper limb movement factors. After determining Pearson r correlation coefficients between the factors and their components, they were assigned appropriate names. Correlation coefficients were also calculated between the determined anthropometric factors, mechanical power, and upper limb movement patterns. The final part of the statistical analysis involved the creation of a correlation matrix (Pearson r coefficients), which took into account anthropometric factors, mechanical power, upper limb movement patterns, and overall kinematic indices. The thresholds adopted were consistent with those previously reported for v and the remaining kinematic variables.

Results

Table 1 presents the mean values ($\pm$ SD) of variables divided into blocks: anthropometric, mechanical power, upper limb movement, and overall technique.

Based on correlation analysis of the overall technique indicators, v was positively associated with SI ($r = 0.71$, 95% CI [0.52; 0.83], $p < 0.01$) and SR ($r = 0.50$, 95% CI [0.22; 0.70], $p < 0.01$). SL, in turn, was positively related to SI ($r = 0.83$, 95% CI [0.70; 0.90], $p < 0.01$) and negatively associated with SR ($r = -0.75$, 95% CI [-0.86; -0.57], $p < 0.01$).

Tables 2, 3 and 4 present the correlation coefficients between the identified factors and their component variables. For the anthropometric variables (Table 2), Factor 1 had an eigenvalue of 6.59, and the percentage of explained variance was 66%. It was named “Large Body Size”. The second anthropometric factor had an eigenvalue of 2.24, and the percentage of explained variance was 23%. Factor 2 was named “Low Fat”.

With respect to the PCA analysis of variables describing mechanical power, one factor was identified with an eigenvalue of 4.42, explaining 74% of the variance. Based on its strong correlations with the component variables (Table 3), this factor was called “Enormous Power”.

Variable category	Variable	Mean $\pm$ SD
Anthropometric	Body height [cm]	163.18 $\pm$ 9.86
	Surface of the hand [mm ²]	931.65 $\pm$ 149.04
	Fold on triceps [mm]	5.61 $\pm$ 1.65
	Fold on shoulder blade [mm]	6.59 $\pm$ 1.94
	Arm span [cm]	167.82 $\pm$ 10.90
	Body mass [kg]	49.86 $\pm$ 10.12
	Body fat [%]	15.94 $\pm$ 2.91
	Active tissue mass of body [kg]	41.87 $\pm$ 8.43
	Active tissue mass of upper limbs [kg]	2.07 $\pm$ 0.56
	Active tissue mass of trunk [kg]	23.43 $\pm$ 4.03
Mechanical power	Peak power [W]	234.76 $\pm$ 70.69
	Relative peak power [W/kg]	4.62 $\pm$ 0.62
	Mean power [W]	203.32 $\pm$ 57.24
	Relative mean power [W/kg]	4.02 $\pm$ 0.48
	Time to reach peak power [s]	7.38 $\pm$ 2.09
	Fatigue Index [%]	9.83 $\pm$ 4.73
Upper limbs' movement	Mean acceleration [m/s ²]	26.83 $\pm$ 4.53
	Maximum acceleration [m/s ²]	68.42 $\pm$ 20.24
	Minimum acceleration [m/s ²]	6.90 $\pm$ 1.98
	Root mean square of acceleration [m/s ²]	30.13 $\pm$ 5.43
	Mean angular velocity [deg/s]	452.06 $\pm$ 53.36
	Maximum angular velocity [deg/s]	1004.5 $\pm$ 187.94
	Minimum angular velocity [deg/s]	113.80 $\pm$ 43.97
	Root mean square of angular velocity [deg/s]	510.82 $\pm$ 60.49
Overall kinematic indices	Velocity [m/s]	1.56 $\pm$ 0.09
	Stroke rate [cycle/min]	53.92 $\pm$ 4.66
	Stroke length [m]	1.74 $\pm$ 0.13
	Stroke index [m ² /s]	2.72 $\pm$ 0.28

Table 1. Means and standard deviations ($\pm$ SD) by category.

Variable	Factor 1 large body size	Factor 2 low fat
Body height	0.95****	0.19
Surface of the hand	0.89***	0.03
Fold on triceps	0.06	-0.89***
Fold on shoulder blade	0.47*	-0.75***
Arm span	0.92****	0.15
Body mass	0.98****	-0.10
Body fat	0.07	-0.89***
Active tissue mass of body	0.98****	0.06
Active tissue mass of upper limbs	0.96****	0.16
Active tissue mass of trunk	0.99****	0.01

Table 2. The value of Pearson's r correlation coefficients between the anthropometric components from the PCA analysis and their variable contributions. *moderate correlation strength, **large correlation strength, ***very large correlation strength, ****nearly perfect correlation strength.

Variable	Variable 1 enormous power
Mean power	0.96****
Relative mean power	0.84***
Peak power	0.97****
Relative peak power	0.90****
Time to reach peak power	-0.75***
Fatigue index	0.71***

Table 3. The value of Pearson's r correlation coefficients between the mechanical power components from the PCA analysis and their variable contributions. *moderate correlation strength, **large correlation strength, ***very large correlation strength, ****nearly perfect correlation strength.

Variable	Factor 1 dynamism	Factor 2 high angular velocity
Mean acceleration	0.94****	-0.20
Maximum acceleration	0.76***	-0.33*
Minimum acceleration	0.43*	-0.34*
RMS acceleration	0.93****	-0.22
Mean angular velocity	0.91****	0.24
Maximum angular velocity	0.63**	0.70***
Minimum angular velocity	0.70***	-0.33*
RMS angular velocity	0.89***	0.41*

Table 4. The value of Pearson's r correlation coefficients between the upper limbs' movement components from the PCA analysis and their variable contributions. *moderate correlation strength, **large correlation strength, ***very large correlation strength, ****nearly perfect correlation strength.

Regarding the variables describing hand movement, two factors were identified with eigenvalues of 5.03 and 1.13, respectively, explaining 63% and 14% of the variance. Factor 1 was labeled Dynamism, whereas Factor 2 was labeled High Angular Velocity (Table 4).

Analysis of the correlation coefficients between the identified factors indicated a large ES for the association between Large Body Size and Enormous Power ($r=0.85$, 95% CI [0.74; 0.92], $p<0.01$). Consistent patterns were also evident for Dynamism and Enormous Power ($r=0.51$, 95% CI [0.22; 0.71], $p<0.01$) as well as for Dynamism and Large Body Size ($r=0.34$, 95% CI [0.04; 0.58], $p=0.03$). The remaining associations between the factors were classified as trivial or small.

Table 5 presents the Pearson correlation coefficients between the overall technique variables and the factors identified in the PCA analysis. Indicator v exhibited moderate-to-large effect sizes, reflected in its associations with Large Body Size ($r=0.56$, 95% CI [0.30; 0.73], $p<0.01$), Enormous Power ($r=0.62$, 95% CI [0.39; 0.78], $p<0.01$), and Dynamism ($r=0.55$, 95% CI [0.29; 0.73], $p<0.01$). A moderate association was also observed between v and Low Fat ($r=0.38$, 95% CI [0.09; 0.60], $p=0.02$).

For SR, the strongest association was observed with Dynamism, corresponding to a large ES ($r=0.76$, 95% CI [0.58; 0.86], $p<0.01$), whereas a weaker pattern emerged in relation to Enormous Power ($r=0.35$, 95% CI [0.05;

Variable	Large body size	Low fat	Enormous power	Dynamism	High angular velocity
velocity	0.56**	0.38*	0.62**	0.55**	-0.12
Stroke Rate	0.17	0.06	0.35*	0.76***	-0.24
Stroke Length	0.23	0.23	0.09	-0.44*	0.19
Stroke Index	0.49*	0.38*	0.42*	0.00	0.06

Table 5. The value of Pearson's r correlation coefficients between the holistic kinematic indices and the factors from PCA analyses. *moderate correlation strength, **large correlation strength, ***very large correlation strength, ****nearly perfect correlation strength.

0.58], $p=0.03$). SL displayed a moderate negative association with Dynamism, consistent with a moderate ES ($r=-0.44$, 95% CI [-0.65; -0.15], $p<0.01$). SI showed moderate effect sizes, expressed through its associations with Large Body Size ($r=0.49$, 95% CI [0.20; 0.69], $p<0.01$), Low Fat ($r=0.38$, 95% CI [0.09; 0.60], $p=0.02$), and Enormous Power ($r=0.42$, 95% CI [0.13; 0.63], $p<0.01$).

Discussion

This research characterized the front crawl technique in young swimmers using IMU sensors, with subsequent data analysis performed partially through AI algorithms. The study results indicate that the overall kinematic variables describing front crawl technique are closely associated with the pattern of arm movement. Nonetheless, the observed effects are additionally modulated by the swimmers' somatic characteristics and the mechanical power of the upper limbs. This study provides a more comprehensive understanding of how young swimmers express their morpho-functional potential through technical performance.

The velocity achieved during full-stroke swimming depends on SL and SR²⁷. These parameters typically exhibit a negative association, which was also observed in the present study ($r=-0.75$). However, the association was not perfect, suggesting that specific combinations of SL and SR may produce an optimal swimming velocity⁷. Such combinations allow athletes to achieve speeds appropriate to their physiological capacities and the demands of a given race distance²⁸. Consequently, a SR–SL profile that is effective in short all-out efforts (25–50 m) may be less advantageous over longer distances, such as 100 m^{28,29}. Therefore, the conclusions from this study should be primarily applied to the 50 m front crawl performance.

Previous research on elite senior and junior swimmers has not demonstrated that velocity is predominantly influenced by either SL or SR⁷. The course of movement of faster swimmers has instead been characterized by slightly higher values of both parameters²⁷. Among swimmers at lower performance levels (including youth categories), velocity over sprint distances has been shown to be more strongly associated with SR than with SL^{8,30}. Our findings align with these observations. Velocity exhibited a moderate ES in relation to SR ($r=0.50$), whereas the corresponding effect for SL was small. The described differences between age groups suggest that the relative contribution of these parameters to swimming velocity may shift with maturation. This likely reflects differences in the individual characteristics underlying of SR and SL⁸. It is generally assumed that the former is determined primarily by mechanical muscle power, while the latter is more closely related to somatic structure^{18,31}. It should be noted that maturation processes proceed at different rates; for example, rapid increases in limb length typically precede substantial gains in muscle mass³². This means that young athletes may temporarily experience difficulty swimming with the high stroke frequency typical of the 50 m distance. These considerations also indicate that combining findings on overall technique parameters across age groups should be interpreted with caution.

In this study, PCA identified two anthropometric components, one component representing upper-limb power variables, and two components related to upper-limb technique. Meaningful effect sizes were observed across all variable groups. The strongest association was observed between Large Body Size and Enormous Power ($r=0.85$), which is consistent with previous research^{8,18,33}. Larger athletes generally possess greater muscle mass, which may, in turn, contribute to superior performance in tests assessing mechanical power^{34,35}. At the same time, the Low Fat component was not associated with Enormous Power. Adipose tissue - often described as metabolically "inactive"³⁶, was therefore not reflected in indicators of mechanical muscle power.

A novel aspect of the study was the examination of how somatic build and mechanical power are related to upper-limb movement patterns described using IMU-derived data. The analysis showed that both the Large Body Size and Enormous Power components were associated with Dynamism, although the magnitude of these effects differed ($r=0.34$ and $r=0.51$, respectively). Variables describing hand acceleration were more strongly related to power than to somatic build. This indicates that the latter association may be partly explained by the aforementioned link between Large Body Size and Enormous Power^{33,34}. The Low Fat component showed no association with Dynamism, supporting the earlier assumptions that adiposity does not substantially contribute to muscle power³⁶. These findings indicate that an athlete's anthropometric and energetic characteristics are reflected in IMU-derived measures and should therefore be considered in future studies employing this technology.

The Dynamism component showed a strong association with selected overall technique parameters - most notably with SR ($r=0.76$), followed by v ($r=0.55$) and SL ($r=-0.44$). This pattern indicates that upper limb acceleration is closely related to effective front crawl sprint performance^{37,38}. The strong correlation coefficient between Dynamism and SR indicates that hand acceleration may be an important characteristic accompanying the SR–SL combination observed during sprint swimming. It is generally assumed that the basis for effective human locomotion in water is, among other things, the ability to accelerate the hand in the appropriate phases of underwater movement^{39,40}. In future technique analyses using IMU technology, it would be reasonable to

record hand acceleration, taking into account the division of movement into underwater and above-water parts. Such detailed characterization may enable a more comprehensive understanding of how hand acceleration is related to SL.

Previous studies describing the determinants of sprint performance have indicated that individuals with high mechanical power and larger body size achieve better results^{31,34,41}. The present findings are consistent with these observations: the associations between velocity and the Large Body Size and Enormous Power components were strong ($r = 0.56$ and 0.62 , respectively). At the same time, Low Fat was moderately related to velocity ($r = 0.38$). In groups with higher performance levels, a negative association between body fat and swimming performance has typically been reported^{33,41}. It is plausible that greater body size - typically accompanied by higher muscle mass - supports achieving high levels of muscle power³⁴. At the same time, excessive body fat may be a factor limiting swimming performance³³. As previously noted, adipose tissue, sometimes described as “inactive,” does not contribute positively to performance in tasks determined by mechanical power³⁶. Excess adiposity may increase the swimmer's cross-sectional area and thereby elevate hydrodynamic drag³⁵. The results of this study complement earlier findings, indicating that the advantage of swimmers with larger bodies and greater mechanical power is revealed by performing upper limb movements characterized by high acceleration. This, in turn, facilitates swimming at higher stroke rates, ultimately enabling athletes to achieve higher velocity⁶. Therefore, interpretations of sprint swimming results in young athletes should account for their stage of biological development. Likewise, body fat levels should be monitored in maturing groups, as they may be related to sprint performance.

It is generally accepted that locomotion in water is significantly less efficient than on land⁴². This is due to the properties of the aquatic environment (high density) and the human body (small propulsive surfaces, body shape promoting large hydrodynamic drag)^{37,43}. This research indicates a possible direction for using IMU sensors to better understand how swimmers can utilize their anthropometric-energy potential in water. Swimmers with sufficiently high mechanical power of the upper limbs are able to perform arm movements characterized by substantial acceleration, thus enabling the attainment of the high SR required to reach significant swimming velocity. The processing of IMU data using AI, as applied in this study, can be used in future research examining the influence of the energy–anthropometric profile on overall kinematic indicators. The SR values determined in this study were compared with those obtained from video analysis, revealing a small difference (1.06%). Given that swimmers with a lower technical level typically demonstrate greater technique variability⁴⁴, similarly small differences may be observed in studies involving older, more advanced athletes.

The main limitation of this study was the lack of separation of IMU data into underwater and above-water phases of the movement. In addition, the measurements primarily focused on hand motion, without considering the contribution of other propulsive surfaces (forearm, upper arm, and feet). In sprint swimming, lower-limb action may contribute substantially to propulsion and influence body position and drag; therefore, the present findings should be interpreted as specific to upper-limb-driven aspects of propulsion. Moreover, test–retest reliability (ICC) of the arm-cranking protocol was not assessed, as each participant performed a single maximal trial. An additional limitation is that the present analysis examined linear associations between PCA-derived components; therefore, the existence of potential threshold effects or non-linear relationships - such as minimum or optimal ranges of anthropometric, power-related, or dynamism-related characteristics - cannot be excluded. The results could also be affected by the stage of biological development, which was not taken into account in this research. Another limitation is that the movement cycles recorded by the IMU sensor may not have fully corresponded to those identified using video analysis. Although the AI-processed data were thoroughly verified by the research team to ensure accuracy, this procedure improved data quality at the expense of substantially increased processing time.

Future research should aim for a more comprehensive analysis of the technique by incorporating movement patterns of other propulsive surfaces. This could be achieved by placing IMU sensors on different body segments. With regard to hand motion, separating underwater and above-water phases may allow a more precise characterization of how hand acceleration is related to SL. The use of Machine Learning mechanisms in data processing could also substantially reduce analysis time, thereby facilitating studies involving larger participant samples.

Conclusions

Based on the results of this study, the following conclusions can be drawn:

1. Overall kinematic parameters characterizing the front crawl technique are associated with variables describing arm movement acceleration and angular velocity.
2. During maturation, young athletes may experience temporary difficulties in sustaining high Stroke Rate, which may limit performance in short-distance swimming events.
3. A comprehensive assessment of swimming technique should consider not only swimming velocity but also Stroke Length and Stroke Rate. Monitoring these parameters enables the evaluation of whether an athlete is optimally utilizing their anthropometric-energetic potential during swimming.
4. Optimization of swimming technique in developing athletes should not focus solely on movement parameters, but should also account for individual characteristics, including somatic build and the capacity to generate mechanical power. High stroke rates appear to be achievable and effective only when accompanied by sufficient anthropometric and power-related potential, whereas attempts to increase stroke rate in swimmers lacking these characteristics may not result in the desired performance improvements.
5. Technique analyses employing inertial measurement unit technology and artificial intelligence facilitate a better understanding of the relationship between anthropometric-energetic potential and overall swimming performance indicators.

Data availability

The datasets generated and analysed during the current study are available in the RODBUK repository, [<http://akf.rodruk.pl/dataset.xhtml?persistentId=doi:10.58145/AKF/ZPKILP>](<https://akf.rodruk.pl/dataset.xhtml?persistentId=doi:10.58145/AKF/ZPKILP>).

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Author contributions

Ł.W. conceptualized the study, conducted the literature review, collected the data, carried out statistical analyses and interpretation of data, and wrote and edited the original draft of the manuscript. R.S. conceptualized the study, carried out statistical analyses and interpretation of data and wrote the draft of the manuscript. M.S. collected the data, performed analysis and interpretation and carried out statistical analyses. K.S. conducted the literature review and edited the draft of the manuscript. J.B. conceptualized the study and reviewed and edited the draft of the manuscript. M.Ż. conceptualized the study and reviewed and edited the draft of the manuscript. Ł.K. conceptualized the study and reviewed and edited the draft of the manuscript. Each author reviewed and approved the submitted version of the manuscript and agreed to be personally accountable for their own contributions and to ensure that questions related to the accuracy or integrity of any part of the work, even ones in which the author was not personally involved, are appropriately investigated, resolved, and the resolution documented in the literature.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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